INCH-POUND

MS39324C
20 January 2017
SUPERSEDING
MS39324B
25 June 2015

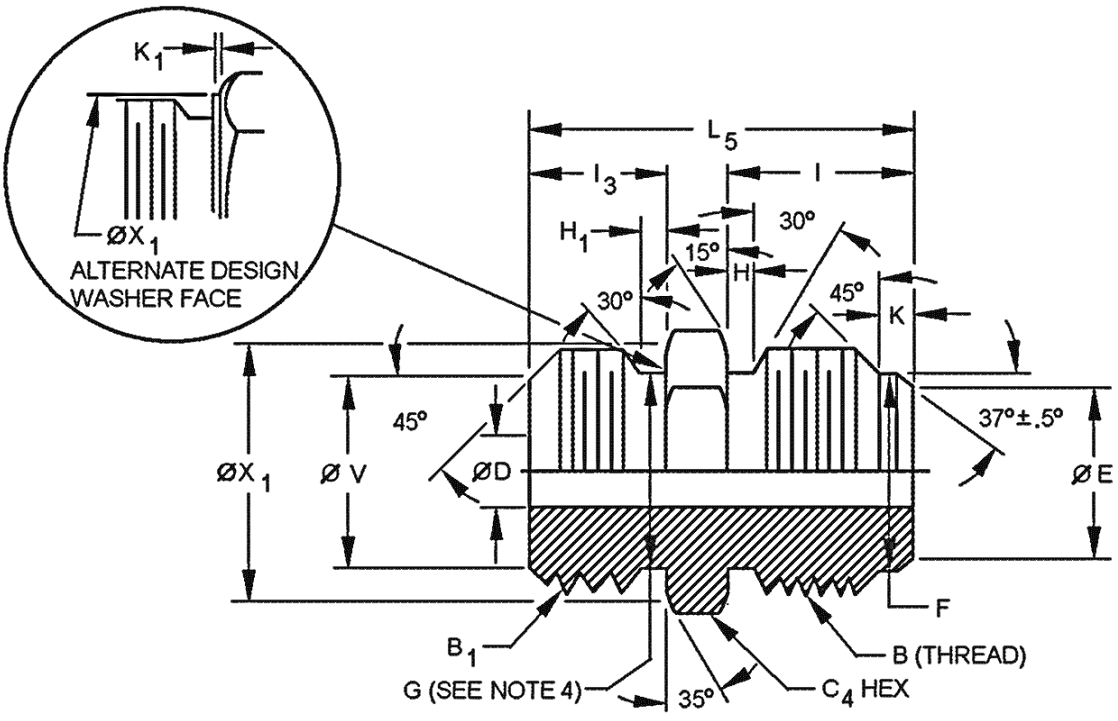
DETAIL SPECIFICATION SHEET

ADAPTER, 37 DEGREE, FLARE TO "O" RING BOSS, 45 DEGREE FLARE

Reactivated after 20 January 2017 and may be used for new and existing designs and acquisitions.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-18666.



See notes at end of figure.

FIGURE 1. Adapter, 37°, flare to "O" ring boss.

AMSC N/A

FSC 4730

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Dash number	Tube O.D. nom	B Straight thread Nom size (see note 5)	B_1 Straight Thread Nom size (see note 5)	C_4	D dia.		E dia.	
				Hex Nom	Inches ±.003	mm ±0.08	Inches ±.003	mm ±0.08
-4-4	1/4 -1/4	.4375-20 UNF-2A	.4375-20 UNF-2A	9/16	.172	4.37	.193	4.90
-5-4	5/16 - 1/4	.5000-20 UNF-2A	.4375-20 UNF-2A	9/16	.172	4.37	.255	6.48
-6-4	3/8 - 1/4	.5625-18 UNF-2A	.4375-20 UNF-2A	5/8	.172	4.37	.318	8.08
-8-4	1/2 - 1/4	.7500-16 UNF-2A	.4375-20 UNF-2A	13/16	.172	4.37	.426	10.82
-10-4	5/8 - 1/4	.8750-14 UNF-2A	.4375-20 UNF-2A	15/16	.172	4.37	.539	13.69
-12-4	3/4 - 1/4	1.0625-12 UN-2A	.4375-20 UNF-2A	1 1/8	.172	4.37	.664	16.87
-5-5	5/16 - 5/16	.5000-20 UNF-2A	.4375-20 UNF-2A	5/8	.234	5.94	.255	6.48
-6-5	3/8 - 5/6	.5625-18 UNF-2A	.5000-20 UNF-2A	5/8	.234	5.94	.318	8.08
-8-5	1/2 - 5/16	.7500-16 UNF-2A	.5000-20 UNF-2A	13/16	.234	5.94	.426	10.82
-10-5	5/8 - 5/16	.8750-14 UNF-2A	.5000-20 UNF-2A	15/16	.234	5.94	.539	13.69
-12-5	3/4 - 5/16	1.0625-12 UN-2A	.5000-20 UNF-2A	1 1/8	.234	5.94	.664	16.87
-6-6	3/8 - 3/8	.5625-18 UNF-2A	.5625-18 UNF-2A	11/16	.297	7.54	.318	8.08
-8-6	1/2 - 3/8	.7500-16 UNF-2A	.5625-18 UNF-2A	13/16	.297	7.54	.426	10.82
-10-6	5/8 - 3/8	.8750-14 UNF-2A	.5625-18 UNF-2A	15/16	.297	7.54	.539	13.69
-12-6	3/4 - 3/8	1.0625-12 UN-2A	.5625-18 UNF-2A	1 1/8	.297	7.54	.664	16.87
-8-8	1/2 - 1/2	.7500-16 UNF-2A	.7500-16 UNF-2A	7/8	.391	9.93	.426	10.82
-10-8	5/8 - 1/2	.8750-14 UNF-2A	.7500-16 UNF-2A	1 1/8	.391	9.93	.539	13.69
-12-8	3/4 - 1/2	1.0625-12 UN-2A	.7500-16 UNF-2A	1 1/4	.391	9.93	.664	16.87
-10-10	5/8 - 5/8	.8750-14 UNF-2A	.8750-14 UNF-2A	1	.484	12.29	.539	13.69
-12-10	3/4 - 5/8	1.0625-12 UN-2A	.8750-14 UNF-2A	1 1/8	.484	12.29	.664	16.87
-12-12	3/4 -3/4	1.0625-12 UN-2A	1.0625-12 UN-2A	1 1/4	.609	15.47	.664	16.87
-12-16	3/4 - 1	1.0625-12 UN-2A	1.3125-12 UN-2A	1 1/2	.609	15.47	.664	16.87

See notes at end of figure.

FIGURE 1. Adapter, 37°, flare to “O” ring boss - Continued.

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Dash number	F		G		H		H_1	
	Inches ±.005	mm ±0.13	Inches ±.002	mm ±0.05	Inches +.015 -.000	mm +0.38 -0.00	Inches +.015 -.000	mm +0.38 -0.00
-4-4	.359	9.12	.364	9.25	.075	1.91	.075	1.91
-5-4	.421	10.69	.364	9.25	.075	1.91	.075	1.91
-6-4	.476	12.09	.364	9.25	.083	2.11	.075	1.91
-8-4	.654	16.61	.364	9.25	.094	2.39	.075	1.91
-10-4	.767	19.48	.364	9.25	.107	2.72	.075	1.91
-12-4	.938	23.83	.364	9.25	.125	3.18	.075	1.91
-5-5	.421	10.69	.427	10.85	.075	1.91	.075	1.91
-6-5	.476	12.09	.427	10.85	.083	2.11	.075	1.91
-8-5	.654	16.61	.427	10.85	.094	2.39	.075	1.91
-10-5	.767	19.48	.427	10.85	.107	2.72	.075	1.91
-12-5	.938	23.83	.427	10.85	.125	3.18	.075	1.91
-6-6	.476	12.09	.482	12.24	.083	2.11	.083	2.11
-8-6	.654	16.61	.482	12.24	.094	2.39	.083	2.11
-10-6	.767	19.48	.482	12.24	.107	2.72	.083	2.11
-12-6	.938	23.83	.482	12.24	.125	3.18	.083	2.11
-8-8	.654	16.61	.660	16.76	.094	2.39	.094	2.39
-10-8	.767	19.48	.660	16.76	.107	2.72	.094	2.39
-12-8	.938	23.83	.660	16.76	.125	3.18	.094	2.39
-10-10	.767	19.48	.773	19.63	.107	2.72	.107	2.72
-12-10	.938	23.83	.773	19.63	.125	3.18	.107	2.72
-12-12	.938	23.83	.945	24.00	.125	3.18	.125	3.18
-12-16	.938	23.83	1.195	30.35	.125	3.18	.125	3.18

See notes at end of figure.

FIGURE 1. Adapter, 37°, flare to “O” ring boss - Continued.

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Dash number	I		I_3		K		K_1	
	Inches ±.005	mm ±0.13	Inches ±.005	mm ±0.13	Inches +.015 -.000	mm +0.38 -0.00	Inches +.015 -.000	mm +0.38 -0.00
-4-4	.550	13.97	.360	9.14	.193	4.90	---	---
-5-4	.550	13.97	.360	9.14	.193	4.90	---	---
-6-4	.556	14.12	.360	9.14	.198	5.03	.062	1.57
-8-4	.657	16.69	.360	9.14	.253	6.43	.062	1.57
-10-4	.758	19.25	.360	9.14	.266	6.76	.062	1.57
-12-4	.864	21.95	.360	9.14	.315	8.00	.062	1.57
-5-5	.550	13.97	.360	9.14	.193	4.90	---	---
-6-5	.556	14.12	.360	9.14	.198	5.03	---	---
-8-5	.657	16.69	.360	9.14	.253	6.43	.062	1.57
-10-5	.758	19.25	.360	9.14	.266	6.76	.062	1.57
-12-5	.864	21.95	.360	9.14	.315	8.00	.062	1.57
-6-6	.556	14.12	.391	9.93	.198	5.03	---	---
-8-6	.657	16.69	.391	9.93	.253	6.43	.062	1.57
-10-6	.758	19.25	.391	9.93	.266	6.76	.062	1.57
-12-6	.864	21.95	.391	9.93	.315	8.00	.062	1.57
-8-8	.657	16.69	.438	11.13	.253	6.43	---	---
-10-8	.758	19.25	.438	11.13	.266	6.76	.094	2.39
-12-8	.864	21.95	.438	11.13	.315	8.00	.094	2.39
-10-10	.758	19.25	.500	12.70	.266	6.76	---	---
-12-10	.864	21.95	.500	12.70	.315	8.00	---	---
-12-12	.864	21.95	.594	15.09	.315	8.00	---	---
-12-16	.864	21.95	.594	15.09	.315	8.00	---	---

See notes at end of figure.

FIGURE 1. Adapter, 37°, flare to “O” ring boss - Continued.

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Dash number	L_5		V dia.		X_1 dia	
	Inches ±.002	mm ±0.05	Inches ±.005	mm ±0.13	Inches +.015 -.000	mm +0.38 -0.00
-4-4	1.23	31.24	.365	9.27	.563	14.30
-5-4	1.23	31.24	.365	9.27	.563	14.30
-6-4	1.23	31.24	.365	9.27	.563	14.30
-8-4	1.34	34.04	.365	9.27	.563	14.30
-10-4	1.47	37.34	.365	9.27	.563	14.30
-12-4	1.64	41.66	.365	9.27	.563	14.30
-5-5	1.23	31.24	.425	10.80	.625	15.88
-6-5	1.23	31.24	.425	10.80	.625	15.88
-8-5	1.34	34.04	.425)	10.80	.625	15.88
-10-5	1.47	37.34	.425	10.80	.625	15.88
-12-5	1.64	41.66	.425	10.80	.625	15.88
-6-6	1.29	32.77	.480	12.19	.688	17.48
-8-6	1.40	35.56	.480	12.19	.688	17.48
-10-6	1.50	38.10	.480	12.19	.688	17.48
-12-6	1.67	42.42	.480	12.19	.688	17.48
-8-8	1.48	37.59	.660	16.76	.875	22.23
-10-8	1.58	40.13	.660	16.76	.875	22.23
-12-8	1.72	43.69	.660	16.76	.875	22.23
-10-10	1.70	43.18	.775	19.69	1.000	25.40
-12-10	1.80	45.72	.775	19.69	1.000	25.40
-12-12	1.97	50.04	.945	24.00	1.250	31.75
-12-16	2.00	50.80	1.195	30.35	1.250	31.75

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Break all sharp edges and remove all burrs and slivers.
4. “O” ring groove shall be smooth and free from tool marks 100µ inches (2.54 µm) maximum.
5. Threads in accordance with ASME B1.1.

FIGURE 1. Adapter, 37°, flare to “O” ring boss - Continued.

REQUIREMENTS:

Fittings shall be as specified on figure 1 and in table I.

Materials shall be in accordance with MIL-DTL-18866.

Finish. Finishes shall be in accordance with table I. All plating’s shall be capable of meeting 96-hour salt spray test in accordance with ASTM B117. The fittings shall show no evidence of red corrosion after 96 hours of salt spray. Fluid passages, other openings, and internal threads shall not be subject to the plating thickness requirement and may have bare areas provided they are protected with a light film of oil.

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TABLE I. Material and finish identification codes.

PIN code material/plating finish	Material	Plating finish
Blank	Steel	Cadmium plating in accordance with SAE-AMS-C-81562, type II, class 3 or SAE-AMS-QQ-P-416, type II, class 2. 1/
H	Steel	Zinc-Aluminum in accordance with ASTM F1136/F1136M, grade 3, NC.
J	Steel	Zinc-nickel in accordance with SAE-AMS2417, type 2, grade B. 2/
M	Nickel-copper alloy UNS N04400	No additional finish.
N	High-chromium nickel alloy UNS N06690	No additional finish.
P	Steel	Zinc phosphate finish in accordance MIL-DTL-16232 type Z, class 1. 3/ 4/
R	Steel	Zinc plating in accordance with ASTM B633; type III or V, Fe/Zn 8. 4/
S	Corrosion resistant steel	No additional finish. Passivation in accordance with SAE-AMS2700, method 1, type 6 or 7.
T	Titanium	Anodize in accordance with SAE-AMS2488 type 2. 5/
TF	Titanium	Fluoride phosphate in accordance with SAE-AMS2486. 5/
Z	Steel	Zinc plating in accordance with ASTM B695, type II, class 8. 4/
ZC	Steel	Zinc may be any zinc plating from PIN codes J, R, and Z with a colored chromate coating. 4/

1/ Embrittlement test need not be run. Cadmium shall not be used in oxygen or potable water systems.
2/ The zinc-nickel alloy plate shall contain 12% to 16% nickel. The coating thickness shall be 315µ inches (8µm) minimum.
3/ Hexavalent chromium free. Finish shall be ROHS compliant.
4/ Not for use in aircraft. Requires approval from the Program Office for all applications.
5/ A pretreatment, a modification of the fluoride treatment, or a post treatment shall be applied so the final color of the fittings shall be similar to FED-STD-595 colors 36076 through 36293. Titanium shall not be used in oxygen or potable water systems.

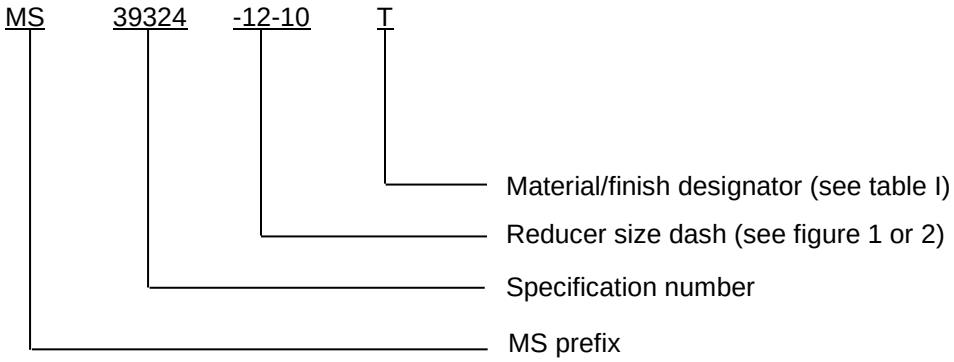
Trivalent wrenchability. When the finish has been damaged due to poor wrenchability, the surface of the connector shall be touched up using one of the brush plating processes below as appropriate to primary finish. The term “trivalent wrenchability” is used to evaluate the ability of the finish to withstand abrasion from an excessive amount of wrenching.

a. Brush plating of hard chromium by electrodeposition shall be in accordance with SAE-AMS-2451/5.
b. Brush plating of medium-hardness, low stress nickel by electrodeposition shall be in accordance with SAE-AMS-2451/9.

Maximum operating pressure. Maximum operating pressure shall be in accordance with SAE-J514.

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PIN: The PIN consists of the letters “MS”, the specification number, a dash, number for adapter size, and a letter for material finish designator.



PIN example: MS39324-12-10T indicates an adapter 3/4 to 5/8 inch (19.05 to 15.88 mm), titanium anodized.

Cadmium is not recommended. To the users of this document, it is recommended that the use of carbon steel material with cadmium plating be used only when other materials and finishes specified in this document cannot meet performance requirements.

Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Referenced documents shall be of the issue in effect on date of invitations for bid.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extent of the changes.

Referenced documents. In addition to MIL-DTL-18666, this document references the following:

FED-STD-595/36076	FED-STD-595/36173	MIL-DTL-16232	SAE-AMS-QQ-P-416
FED-STD-595/36081	FED-STD-595/36176	ASME B1.1	SAE-AMS2417
FED-STD-595/36099	FED-STD-595/36231	ASTM B117	SAE-AMS2451/5
FED-STD-595/36118	FED-STD-595/36251	ASTM B633	SAE-AMS2451/9
FED-STD-595/36134	FED-STD-595/36270	ASTM B695	SAE-AMS2486
FED-STD-595/36152	FED-STD-595/36280	ASTM F1136/F1136M	SAE-AMS2488
FED-STD-595/36170	FED-STD-595/36293	SAE-AMS-C-81562	SAE-AMS2700
			SAE-J514

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CONCLUDING MATERIAL

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Navy - SH
Air Force - 99
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Preparing activity:
DLA - CC

(Project 4730-2017-004)

Review activities:
Army - AR, MI
Navy - CG, MC, SA
Air Force - 71

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